

KausalyaDrishti (ಕೌಶಲ್ಯ ದೃಷ್ಟಿ)

AI-Based Real-Time Monitoring of Training Centres for Attendance and Infrastructure Compliance

PROBLEM STATEMENT IN ONE LINE

Ministry of Skill Development (PS 26245): Government skilling schemes (PMKVY, DDU-GKY) suffer from attendance falsification (ghost trainees claimed for government stipends) and non-compliant lab infrastructure between manual periodic audits.

OUR SOLUTION IN ONE LINE

KausalyaDrishti is an **edge-AI video compliance system** that processes training centre camera feeds in real time to count actual students, detect mandatory machinery, blur faces for 100% trainee privacy, and flag discrepancies to an MSDE ministry command dashboard using only 5 KB of rural bandwidth.

Core Deliverables & PS Requirements Mapping

OFFICIAL REQUIREMENT	KAUSALYADRISHTI IMPLEMENTATION
Real-Time Attendance Estimation	Client-side AI headcount detector continuously counting trainees in classroom video feeds.
Cross-Checking with Portal Records	Tally engine comparing centre's portal claims vs AI headcount → Flags Ghost Trainee Fraud.
Infrastructure Presence & Operability	Computer vision object detector scanning for sanctioned workbenches, sewing machines, welding kits.
Central Discrepancy Dashboard	MSDE Ministry dashboard with district heatmaps, center compliance ratings, and grant freeze actions.
Trainee Privacy Preservation	Automatic dynamic face pixelation/blurring filter conforming to DPDP Act standards.
Low-Bandwidth Rural Mode	Transmits 5 KB JSON metadata telemetry + violation keyframe snapshots instead of 24/7 video.

3–Portal System Architecture

Centre Edge Camera Portal

Runs at Training Centre:

Processes webcam/CCTV feed with live bounding box overlays and privacy face-blur.

Audit & Discrepancy Desk

Automated Audit:

Tally comparison between registered roster and live AI count + Machinery checklist.

MSDE Ministry Dashboard

Used by MSDE / NSDC Officers:

National/State risk heatmaps, show-cause notices, grant freezes.